DATA ANALYTICS

FINAL PROJECT REPORT

TOPIC:
Real Estate Analysis with time series.

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Introduction:

- The datasets we have used are zillow_data.csv and manhattan_brooklyn_zip.csv from kaggle.

- The main objective of using the brooklyn dataset is that brooklyn has favoured investors and buyers because of the relatively low price of rentals.

- In this analysis, our goal is to provide 5 zip codes/neighbourhoods that could maximise the ROI(Return of Investment)
  ROI=Selling Price- Buying Price

- The methodology we have used is the Time series model with ARIMA.

- The steps adopted are
  Step 1: Data processing
  Step 2: Data exploration and visualisation
  Step 3: Decide on model approach and KPI
  Step 4: Develop Initial SARIMA model on sample zip codes
  Step 5: Interpreting the results and making future predictions

Step 1: Data processing – transform raw data onto time series with Time being the index

Importing all the necessary libraries

```
import warnings
warnings.filterwarnings('ignore')
warnings.simplefilter("ignore")

import itertools
import pandas as pd
import numpy as np

#for modeling
import statsmodels.api as sm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tsa.stattools import adfuller
from statsmodels.tsa.seasonal import seasonal_decompose as sd
from sklearn.metrics import mean_squared_error
from sklearn.linear_model import LassoLarsCV

#for visualizations
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
from matplotlib.pylab import rcParams
import time
```

```
df = pd.read_csv('../input/zilkkdfd/zillow_data.csv')
df.head()
```

	RegionID	RegionName	City	State	Metro	CountyName	SizeRank	1996-04	1996-05	1996-06	...	2017-07	2017-08	2017-09	2017-10	20
0	84654	60657	Chicago	IL	Chicago	Cook	1	334200.0	335400.0	336500.0	...	1005500	1007500	1007800	1009600	1013
1	90668	75070	McKinney	TX	Dallas-Fort Worth	Collin	2	235700.0	236900.0	236700.0	...	308000	310000	312500	314100	315
2	91982	77494	Katy	TX	Houston	Harris	3	210400.0	212200.0	212200.0	...	321000	320600	320200	320400	320
3	84616	60614	Chicago	IL	Chicago	Cook	4	498100.0	500900.0	503100.0	...	1289800	1287700	1287400	1291500	1296
4	93144	79936	El Paso	TX	El Paso	El Paso	5	77300.0	77300.0	77300.0	...	119100	119400	120000	120300	120

5 rows × 272 columns

```
ny2 = ny1[ny1['District'].notnull()]
ny2.groupby('District').count()
```

District	RegionID	RegionName	City	State	Metro	CountyName	SizeRank	1996-04	1996-05	1996-06	...	2017-07	2017-08	2017-09	2017-10	2017-11	2017-12	201
Brooklyn	29	29	29	29	29	29	29	25	25	25	...	29	29	29	29	29	29	2
Manhattan	4	4	4	4	4	4	4	1	1	1	...	4	4	4	4	4	4	

2 rows × 272 columns

Only 4 zipcodes in Manhattan are present in the database. We will focus on Brooklyn then.

Unprocessed dataset :-

```
ny4.head()
```

	RegionName	1996-04	1996-05	1996-06	1996-07	1996-08	1996-09	1996-10	1996-11	1996-12	...	2017-07	2017-08	2017-09	2017-10	2017-1
1	11226	162000.0	162300.0	162600.0	163000.0	163400.0	164000.0	164700.0	165700.0	166800.0	...	860200	851000	853900	870000	8851
3	11235	190500.0	191000.0	191500.0	192000.0	192500.0	193000.0	193600.0	194300.0	195300.0	...	767300	777300	788800	793900	7960
9	11230	230100.0	230800.0	231500.0	232100.0	232800.0	233600.0	234700.0	235900.0	237500.0	...	1079200	1082700	1082800	1084900	10912
10	11209	255700.0	256600.0	257300.0	258100.0	258900.0	259800.0	260900.0	262000.0	263600.0	...	1005000	1013400	1019500	1024400	10249
11	11212	125000.0	125000.0	125000.0	125000.0	125100.0	125100.0	125000.0	124800.0	124600.0	...	380000	387000	394600	401500	4079

5 rows × 266 columns

Reshaped and concise dataset :- melt here allows reshaping the data from long to wide format

```
def melt_data(df):
    melted = pd.melt(df, id_vars=['RegionName'], var_name='Month', value_name = 'MeanValue')
    melted['Month'] = pd.to_datetime(melted['Month'], format = '%Y-%m')
    melted = melted.dropna(subset=['MeanValue'])
    return melted
```

```
dfm.head()
```

[16]:

	RegionName	Month	MeanValue
0	11226	1996-04-01	162000.0
1	11235	1996-04-01	190500.0
2	11230	1996-04-01	230100.0
3	11209	1996-04-01	255700.0
4	11212	1996-04-01	125000.0

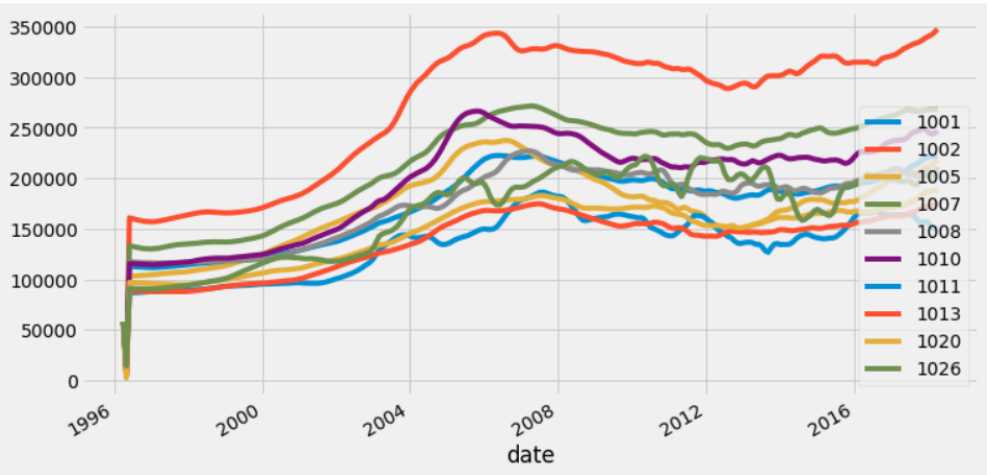
```
#dfm.set_index('Month',inplace=True)
dfm.head(10)
```

	RegionName	MeanValue
Month		
1996-04-01	11226	162000.0
1996-04-01	11235	190500.0
1996-04-01	11230	230100.0
1996-04-01	11209	255700.0
1996-04-01	11212	125000.0
1996-04-01	11214	191000.0
1996-04-01	11221	133700.0
1996-04-01	11229	183900.0
1996-04-01	11234	172300.0
1996-04-01	11215	225700.0

Step 2: Data Exploration and Visualization – Seasonal Decomposition, Autocorrelation with ACF and PACF, etc.

Looking at the overall trend of house prices:

PRICES OF FIRST 10 ZIP CODES (REGION NAMES) OVER TIME

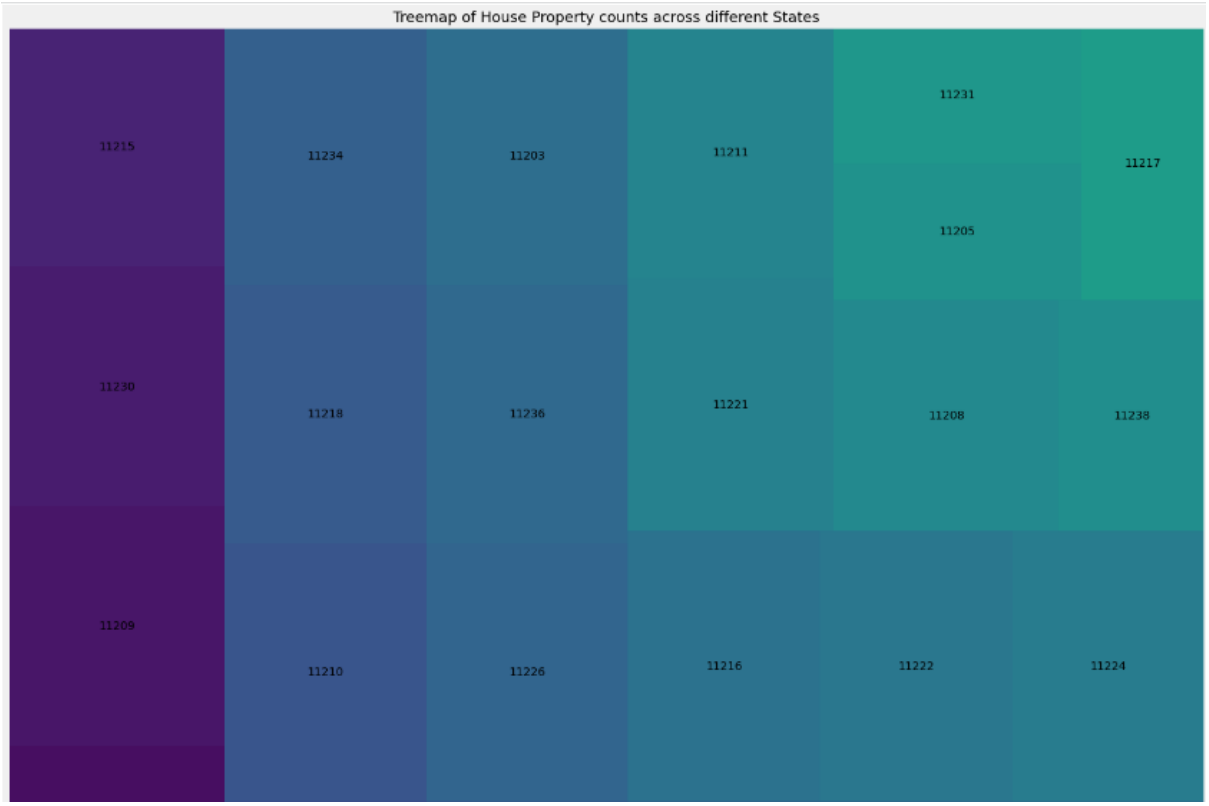


The individual zip codes follow a similar general trend of rising till late 2007 and falling between 2007 and 2010.

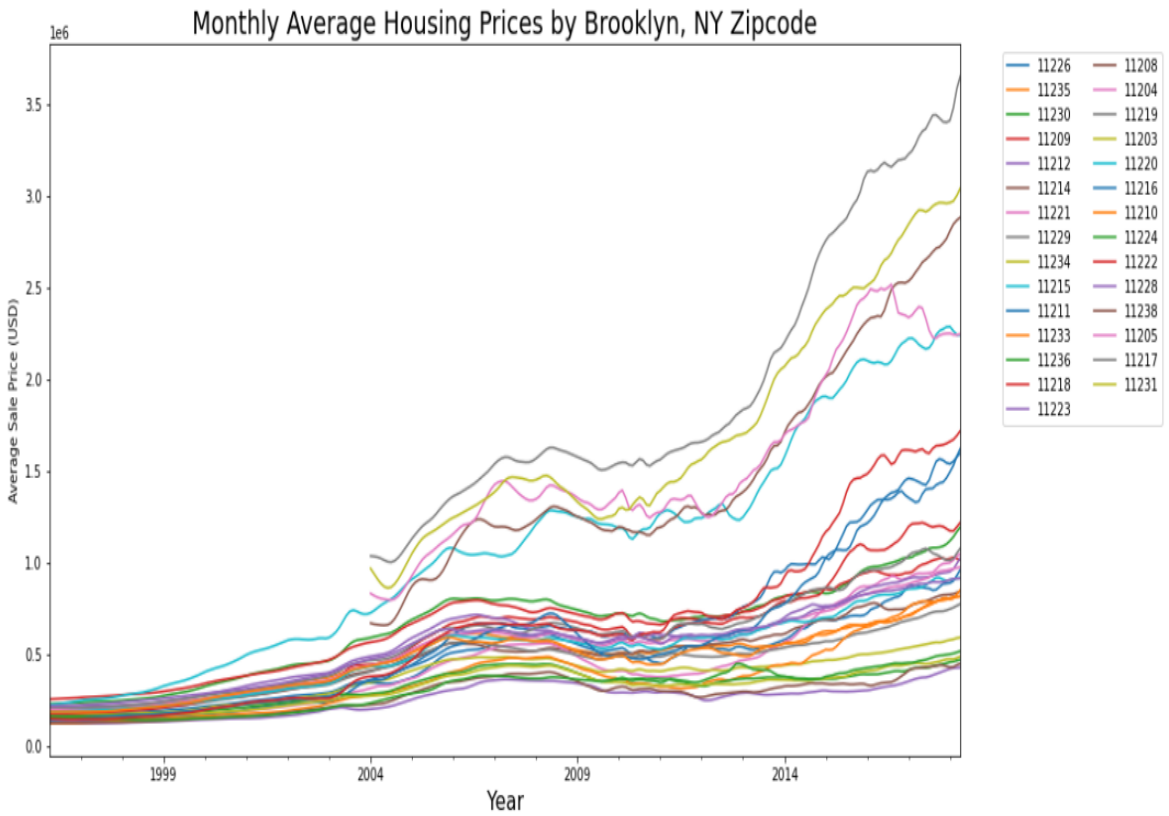
This is observed to be due to the subprime mortgage crisis that crashed the market between 2007 and 2010, and then a subsequent recovery around 2013.

Treemap is a complex representation of hierarchical relationship between attributes, here house price vs zipcodes.

Treemap of house prices vs zipcodes



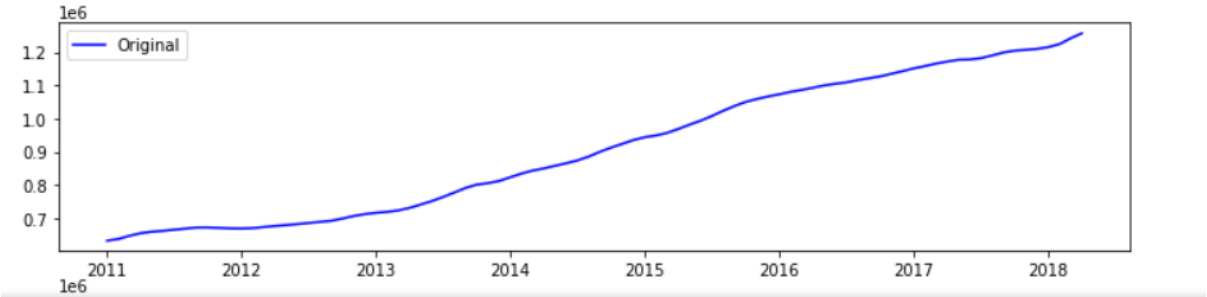
The image below shows the average sales price for each year and how it varies year to year. It can be observed that 25 zip codes have data from 1996 to 2018, 4 zips do not.
There was a dip in 2008 and 2010 because of the financial crisis, housing prices then became stable around 2011.

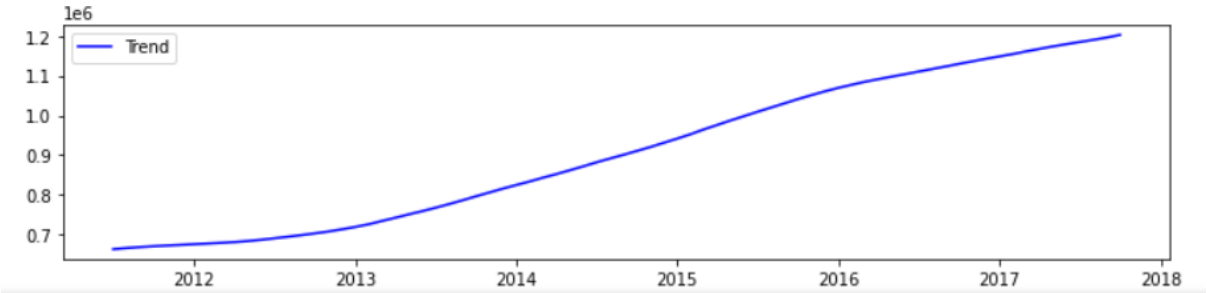


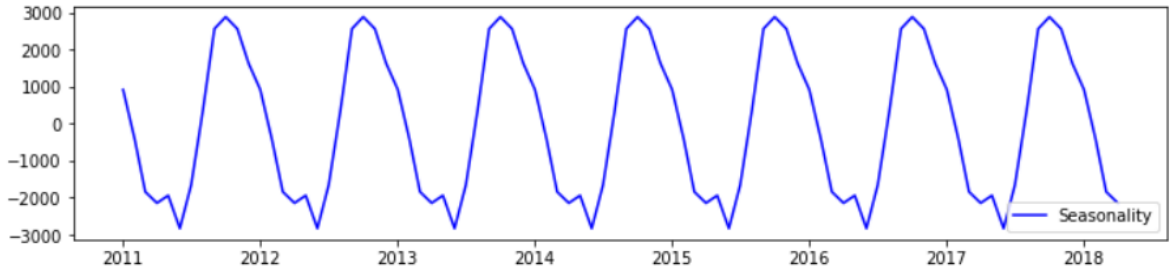
Trend & seasonality decomposition :-

Here we see the seasonality decomposition for 2011 and above.

```
df2011 = dfm['2011':]
month_avg = df2011.groupby('Month')['MeanValue'].mean()
```

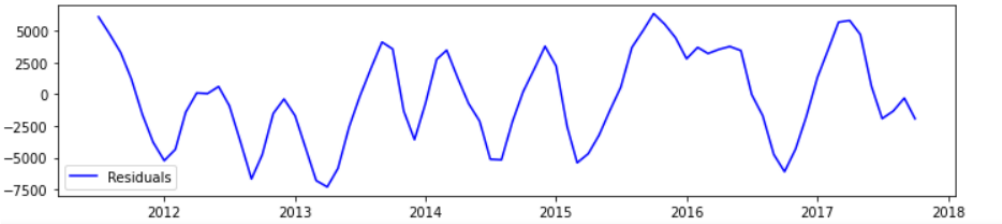






We can observe that an upward trend is present with yearly seasonal changes.

Residuals:-

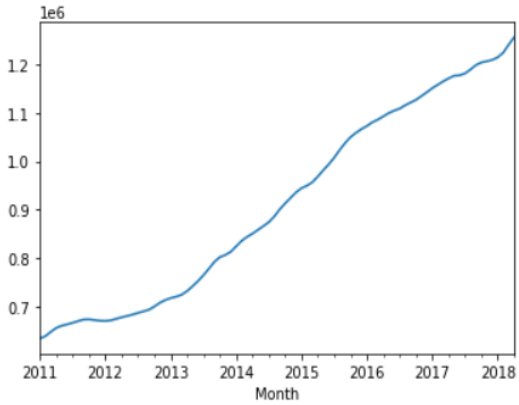


The Dickey Fuller test is used to test stationarity of residuals.

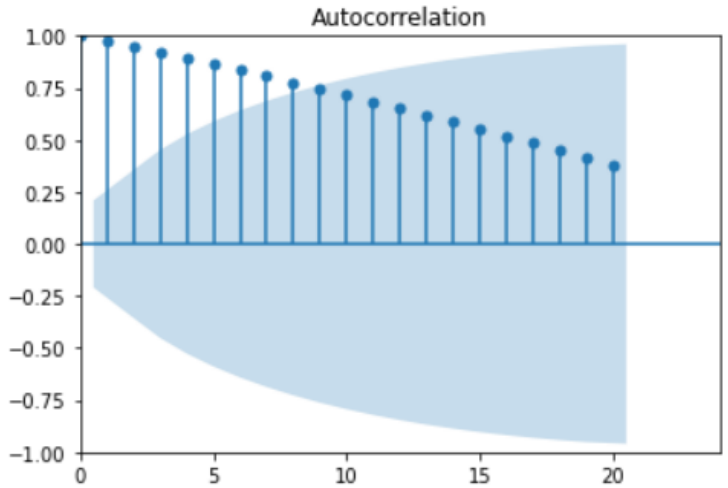
The null hypothesis is the existence of autoregression, and the alternative hypothesis is stationarity.

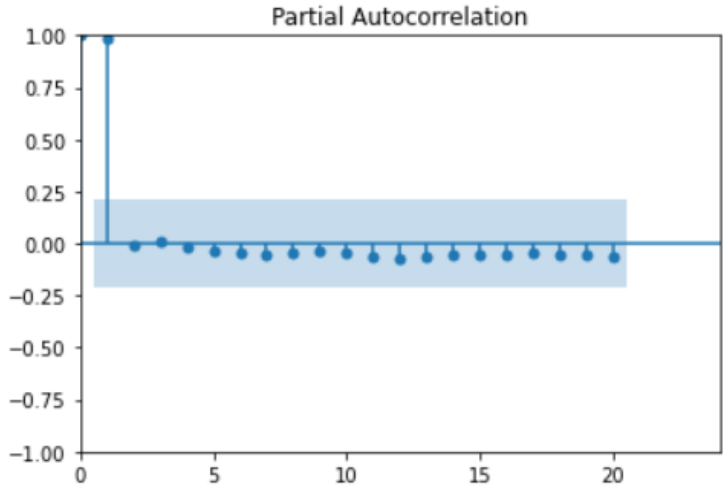
```
Results of Dickey-Fuller Test:

Test Statistic                  -2.527008
p-value                          0.109040
#Lags Used                       6.000000
Number of Observations Used     69.000000
Critical Value (1%)             -3.528890
Critical Value (5%)             -2.904440
Critical Value (10%)            -2.589656
```

Since the p-value is larger than 0.05 we can accept Ho which means the residuals are not stationary.



The monthly average house price of Brooklyn increased year over year. No obvious autocorrelation and partial autocorrelation is present.



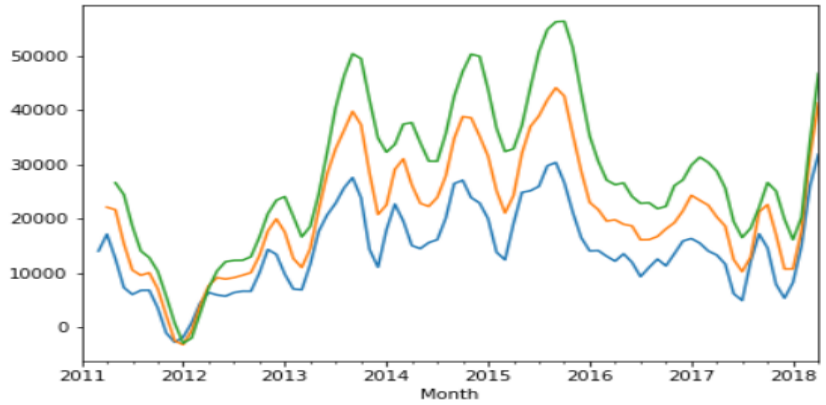


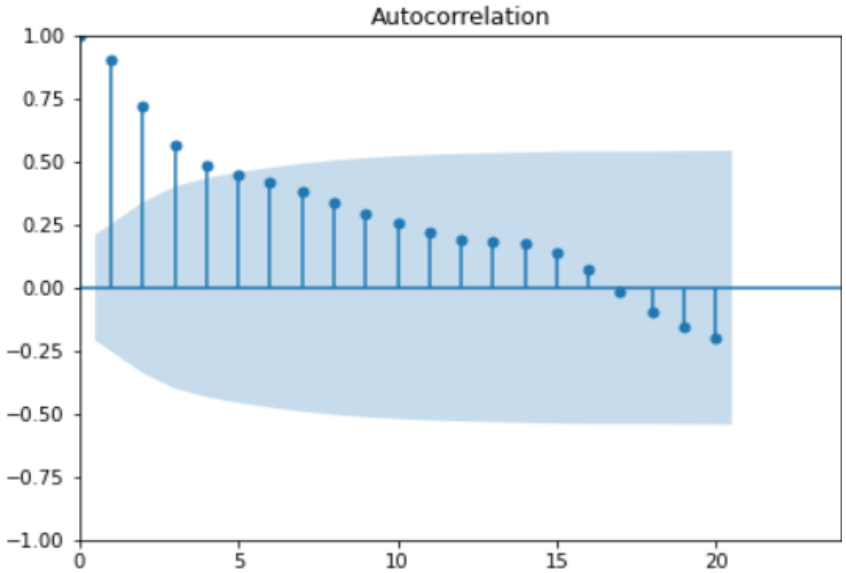
A lag k autocorrelation is the correlation between values that are k time periods apart.

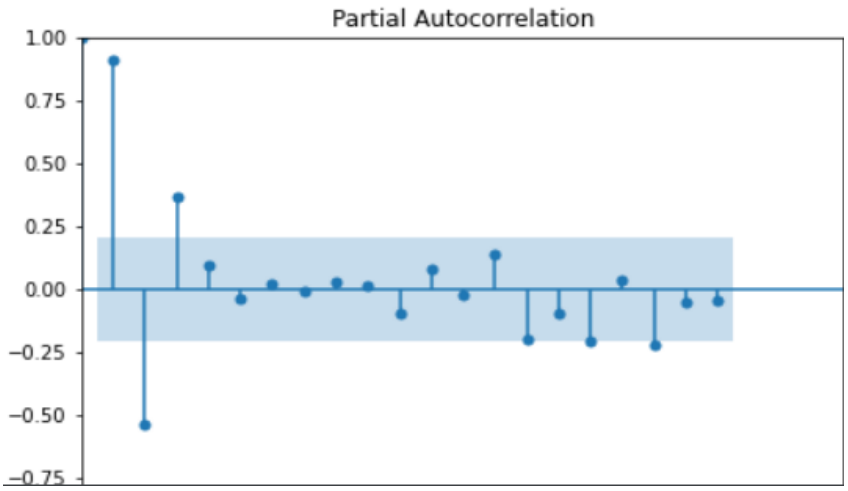
Taking lags :- (we can consider 2 or 3 as difference)

```
month_avg.diff(periods=2).plot()
month_avg.diff(periods=3).plot()
month_avg.diff(periods=4).plot()
```

```
<AxesSubplot:xlabel='Month'>
```





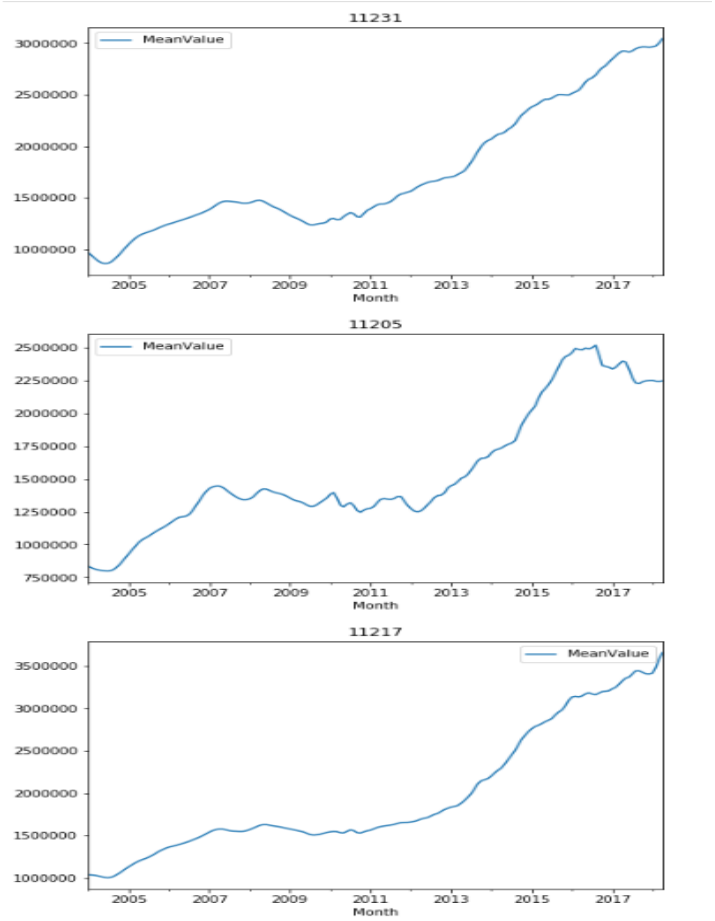


Step 3: Decide on Model Approach and KPI

Since our dataset is not stationary, and there is a seasonal component, it would be reasonable to use the SARIMA model — Seasonal ARIMA (Seasonal Autoregressive Integrated Moving Averages with exogenous regressors).
Key performance indicators are determined by the AIC method.
Akaike Information Criteria (AIC) is widely used to measure a model by quantifying goodness of fit and the simplicity of a model into a simple statistic.
When comparing any two models, the one with lower AIC is the better one.

Step 4.Initial SARIMA Model on Sample Zip Codes

We perform SARIMA on 3 sample zip codes.



Iterate through 0, 1 and 2 for values of p, d, q to find the best fit SARIMAX model.

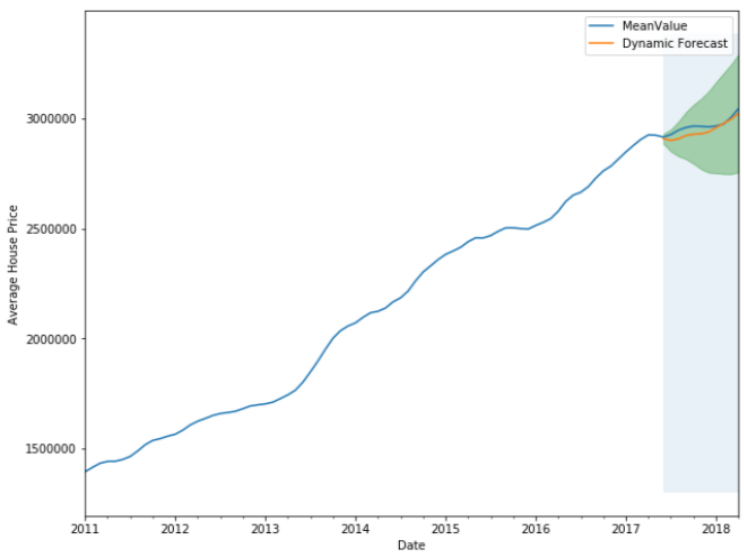
```
print(pdqs)
```

```
[(0, 0, 0, 12), (0, 0, 1, 12), (0, 1, 0, 12), (0, 1, 1, 12), (1, 0, 0, 12), (1, 0, 1, 12), (1, 1, 0, 12), (1, 1, 1, 12)]
```

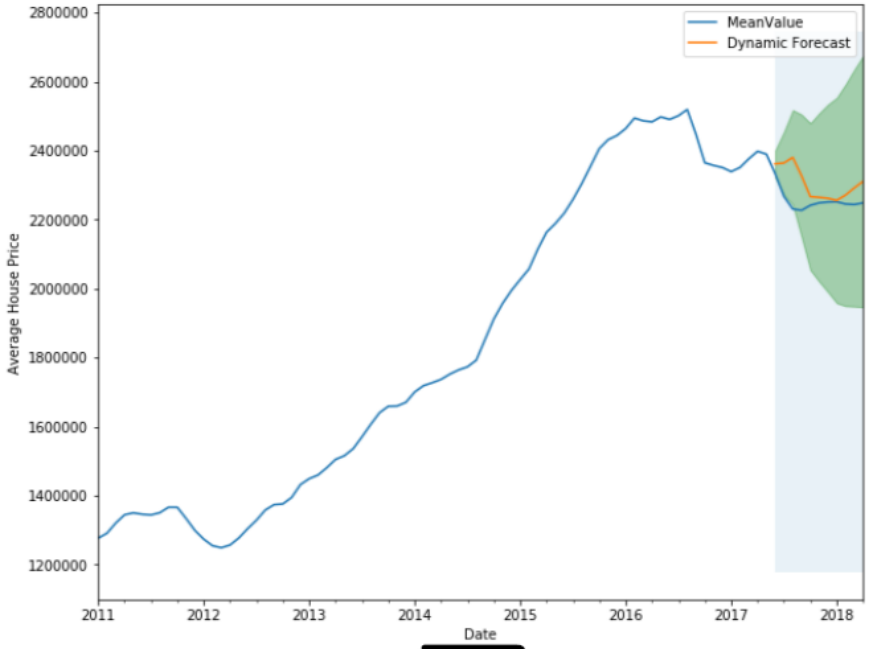
The following zip codes are considered by plugging in the optimal parameters (obtained from AIC) into a new SARIMAX and the summary table is obtained.

	Zipcode	Sqrt_MSE
0	11205	25756.564490
1	11217	67677.540816
2	11231	70471.236940

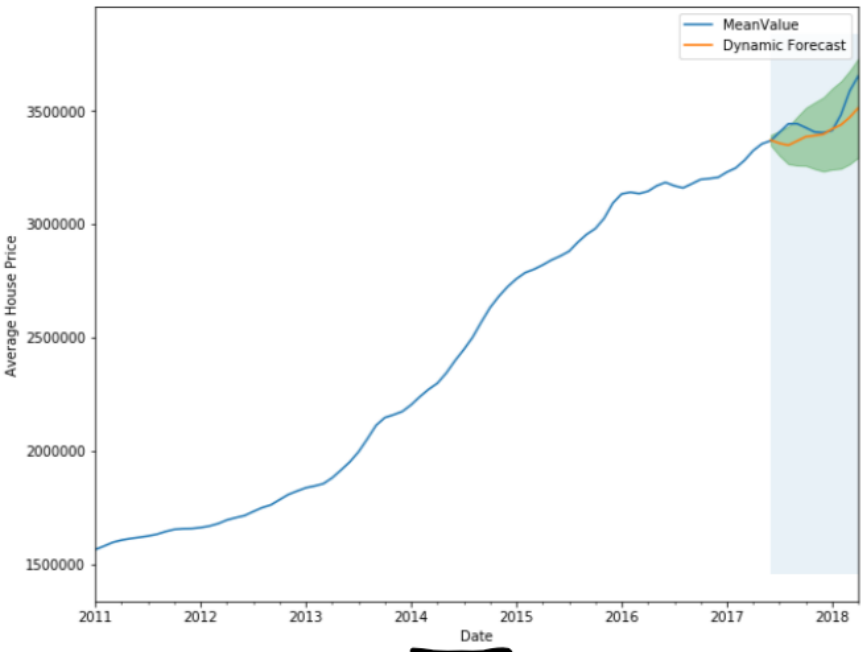
Plotting the dynamic forecasts for each of the above zip codes




```
The Mean Squared Error of our forecasts is 25756.56
```




```
The Mean Squared Error of our forecasts is 67677.54
```




```
The Mean Squared Error of our forecasts is 70471.24
```

We make dynamic forecast on 2017-06-01 onwards and compare with real values, it is found that 5 zip codes that could potentially maximise your Return on Investment in 3 years and 5 years

3yr ROI

```
forecast_table.sort_values('3Yr-ROI', ascending = False)[0:5]
```

	Zipcode	Current Value	3 Years Value	5 Years Value	3 Years Lower	3 Years Upper	5 Years Lower	5 Years Upper	3Yr-ROI
14	11220	1026600.0	1.562324e+06	1.906698e+06	1.203992e+06	1.920656e+06	1.196597e+06	2.616800e+06	0.521843
2	11205	1195400.0	1.765556e+06	2.139029e+06	1.393440e+06	2.137671e+06	1.398717e+06	2.879340e+06	0.476958
0	11203	963200.0	1.379875e+06	1.642406e+06	1.040545e+06	1.719205e+06	1.003086e+06	2.281727e+06	0.432594
18	11224	489900.0	6.746460e+05	7.925614e+05	4.872264e+05	8.620656e+05	4.230197e+05	1.162103e+06	0.377110
11	11217	852300.0	1.138072e+06	1.324995e+06	9.043490e+05	1.371795e+06	8.833630e+05	1.766627e+06	0.335295

5 yr ROI

```
forecast_table.sort_values('5Yr-ROI', ascending = False)[0:5]
```

	Zipcode	Current Value	3 Years Value	5 Years Value	3 Years Lower	3 Years Upper	5 Years Lower	5 Years Upper	3Yr-ROI	5Yr-ROI
14	11220	1026600.0	1.562324e+06	1.906698e+06	1.203992e+06	1.920656e+06	1.196597e+06	2.616800e+06	0.521843	0.857294
2	11205	1195400.0	1.765556e+06	2.139029e+06	1.393440e+06	2.137671e+06	1.398717e+06	2.879340e+06	0.476958	0.789383
0	11203	963200.0	1.379875e+06	1.642406e+06	1.040545e+06	1.719205e+06	1.003086e+06	2.281727e+06	0.432594	0.705156
18	11224	489900.0	6.746460e+05	7.925614e+05	4.872264e+05	8.620656e+05	4.230197e+05	1.162103e+06	0.377110	0.617802
11	11217	852300.0	1.138072e+06	1.324995e+06	9.043490e+05	1.371795e+06	8.833630e+05	1.766627e+06	0.335295	0.554611

Step 5: Interpreting the results and making future predictions

Based on 3 year return on investment and 5 year return on investment, below 5 zipcodes stand out to be the best area to invest in. South Sunset Park, Clinton Hill, East Flatbush and Boerum Hill have a guaranteed positive ROI and could be as high as 87% in 3 years and 154% in 5 years.

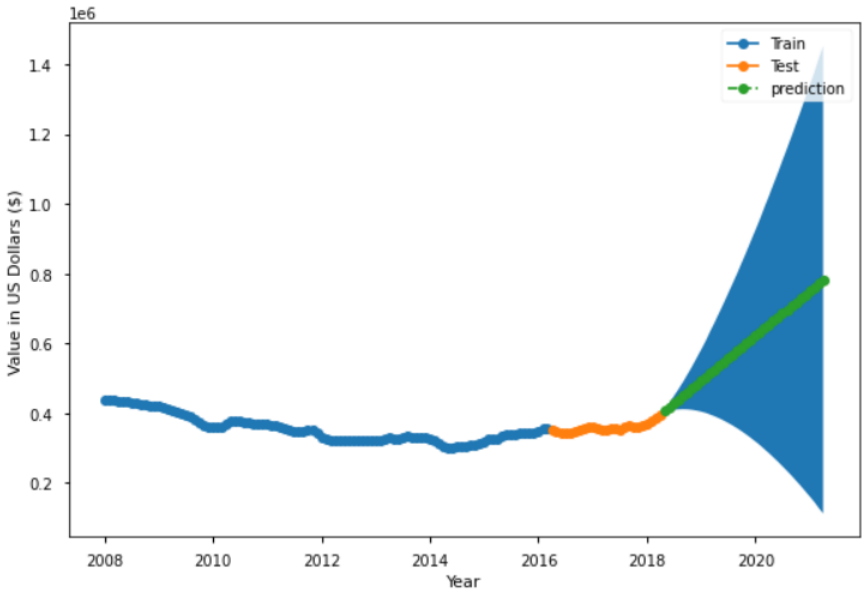
However Coney Island has the risk of decreasing price because it is relatively far from the main city and the value depends on more factors such as development, facilities around the neighbourhood and transportation.

- 11220: South Sunset Park (3Yr ROI: 17%-87%; 5Yr ROI: 16%-154%)
- 11205: Clinton Hill (3Yr ROI: 16%-78%; 5Yr ROI: 17%-140%)
- 11203: East Flatbush (3Yr ROI: 8%-78%; 5Yr ROI: 4%-136%)
- 11224: Coney Island (3Yr ROI: -0.5%-76%; 5Yr ROI: -13%-137%)
- 11217: Boerum Hill (3Yr ROI: 6%-61%; 5Yr ROI: 3%-107%)

However, other factors need to be taken into consideration due to the limit of the time series models, such as transportation, public safety, number of new constructions in the area, and school district etc.

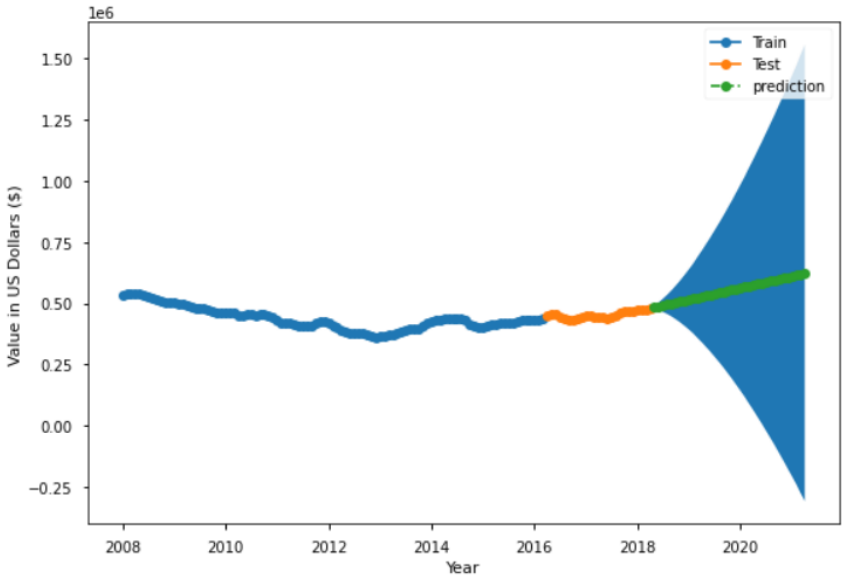
Similarly if we choose Connecticut with ARIMA models of top 3 zip codes:-

- 06069



06069

- 06039



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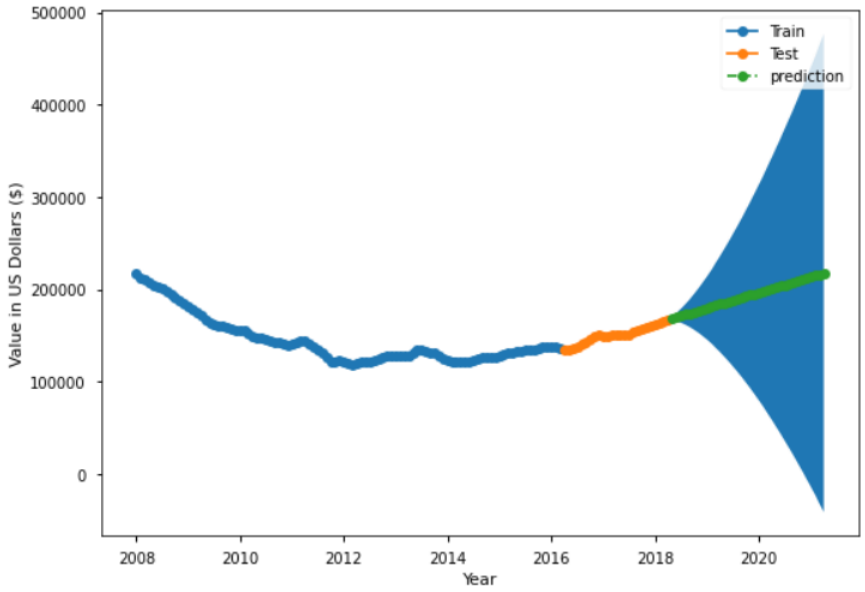
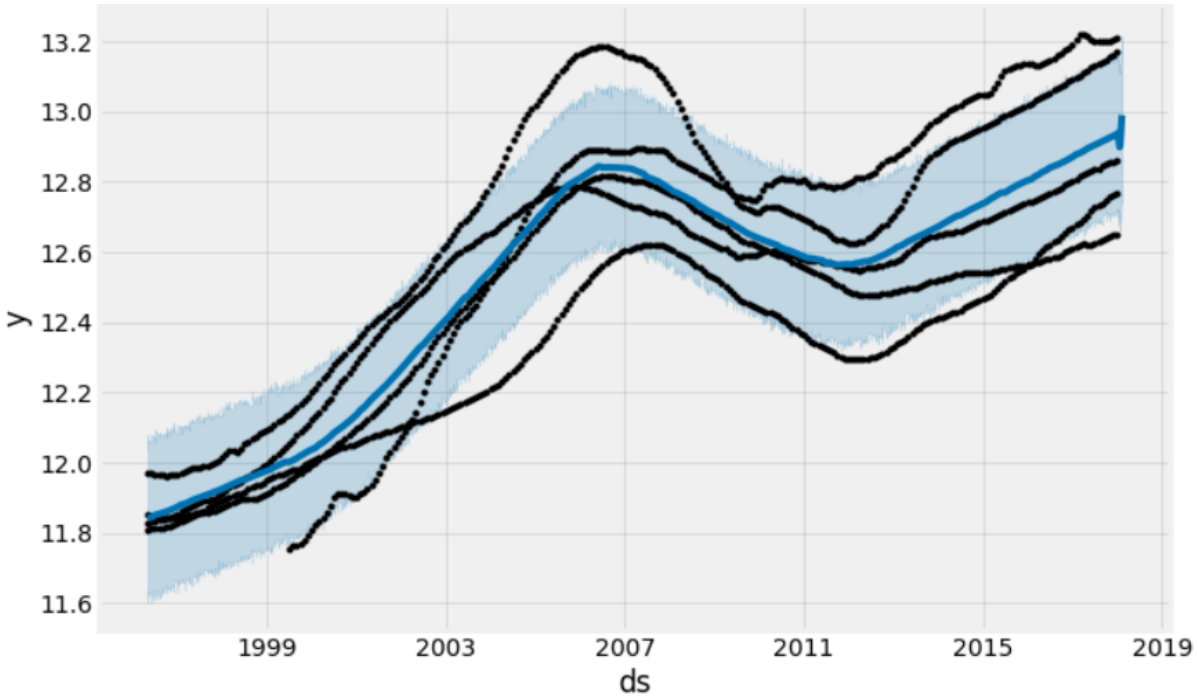


06610

We have also observed the use of Facebook Prophet library on the main house prices master dataset to predict future house prices .